

Localizing and Beating the Out-of-Distribution Bottleneck in Vector Search: A Measured Collection of Routing, Coverage, and Systems Techniques

[TODO: authors]

Abstract

Scalable approximate nearest-neighbor (ANN) search routes a query to a few partition cells, scans quantized codes inside them, and re-ranks a shortlist. On *out-of-distribution* (OOD) workloads — where queries come from a different distribution than the database, as in cross-modal (text→image) retrieval under inner product — we first *localize the bottleneck*: a battery of scan-side and code-side improvements (wider SIMD, higher-resolution and rotated codes, RaBitQ, ADC routing) leave OOD recall/QPS essentially unmoved, while the achievable recall is capped by which cells get *routed to and covered*. We then present a collection of routing- and coverage-directed techniques, each measured in isolation: a one-parameter routing recalibration (γ) grounded in a cap-versus-ball selectivity argument; k -NN-graph coverage augmentation; a joint cross-level partition optimization; multi-assignment; and a corollary that *reverses* standard practice — once graph coverage is present, coarser cells win. Composed and evaluated under a same-hardware, tight-pairwise protocol against ScaNN at its official configuration, the engine attains higher QPS at matched recall@10 at 1M, 10M[[pending: , and 100M](#)], and transfers to the streaming track. We report the negative results as first-class evidence for the localization, and treat the measurement methodology — which repeatedly caught mirages during development — as a contribution.

1 Introduction

[TODO: Motivate OOD vector search (RAG, cross-modal). The IVF+quant pipeline: route→scan→re-rank. Thesis sentence: on OOD the binding constraint is ROUTING/COVERAGE, not scan throughput or code fidelity – and we show this by measuring both sides. Then: a collection of coverage/routing techniques, each with its own measured effect, plus the informative negatives.]

Contributions (each measured in isolation).

- **A bottleneck localization** (§4): scan- and code-side levers do not move OOD; routing/coverage does. The negatives (§11) are the evidence.
- γ **routing recalibration** (§5), grounded in a cap-vs-ball selectivity argument (§3). *Effect: $\approx$ half the probes at matched OOD recall, zero query-time cost.*
- k -**NN-graph coverage** (§6). *Effect: composes with γ ; at 1M moves the ScaNN ratio $1.18 \rightarrow 0.98$.*
- **The coarse-cell corollary** (§7). *Effect: with graph coverage present, coarser cells win; at 10M $1.01 \rightarrow 0.85$.*
- **Joint cross-level partition optimization (tree-EM)** (§8). *Effect: $+0.6$ – 1.0 recall@10 points at 10M, QPS-neutral.*
- **Multi-assignment (SOAR)** (§9). *Effect: raises the routing recall ceiling to baseline parity.*
- **Systems levers for streaming** (§10): batch-parallel insert ($4.5\times$), low-live adaptive p (worst step $0.74 \rightarrow 0.94$), and a parallelization fix ($7.6\times$ at 8 threads).

- **A same-hardware evaluation methodology** (§12) and an honest scope of what the ratios do and do not claim.

2 Background

[TODO: IVF-PQ; anisotropic score-aware quantization (ScaNN AH+LUT16); k -NN graphs (HNSW, DiskANN); residual and exact re-ranking. The OOD track (text2image): image base, text queries, inner product, float ground truth. Base norms concentrated but not constant; queries unit-norm.]

3 Theoretical Grounding (Scoped)

[TODO: (i) Cap-vs-ball under concentration: additive ball filters are non-selective in high dimension (everything within $R + r$), angular/spherical-cap filters remain selective; k -means on centered data realizes a data-dependent angular partition. (ii) This frames γ : the routing score $\gamma\|c\|^2 - 2q \cdot c$ interpolates $\gamma=1$ (additive ball / L_2) to $\gamma \rightarrow 0$ (pure inner product / cap), so the theory predicts $\gamma < 1$ is needed precisely in the concentrated OOD regime – which we confirm. (iii) Two principles we build on: route-precise/rank-cheap, and multi-assignment as a space/time tradeoff. Explicitly state we do NOT claim the recursive-tree polylog result (refuted at $\leq 10M$ scale).]

4 Localizing the OOD Bottleneck

[TODO: Frame the diagnostic: pool-recall ceiling (re-rank the whole probed pool) vs scan-limited recall. Show scan is already extracting the pool (no scan headroom), so recall is set by routing/coverage. This motivates attacking routing (§5-9) and predicts scan/code levers won't help (§11, the negatives).]

5 γ : One-Parameter Routing Recalibration

[TODO: Per-cell additive $i32$ bias $(\gamma-1)\|c\|^2$ precomputed at build; zero per-query cost. Fit once per dataset ($\gamma \approx 0.5$ on text2image, 1M and 10M). Ablate $\gamma \in \{0.4, 0.45, 0.5, 0.55, 0.6\}$.]

Effect. Reaches matched OOD recall at $\approx$ half the probes (p : 54 $\rightarrow$ 29 at 1M; $\approx 2.5\times$ probe cut at 10M). Zero query-time cost.

6 k -NN-Graph Coverage Augmentation

[TODO: Union the graph neighbors of the top- M shortlist members into the exact re-rank set (base-side IP-kNN sidecar; built by ScaNN self-search or engine self-query, $\sim 92\%$ edge agreement). Recovers deep neighbors a shallow probe missed. Composes additively with γ (partial overlap, compound ~ 0.97).]

Effect. At 1M, composed with γ , the ScaNN ratio drops 1.18 $\rightarrow$ 0.98 (the first sub-1 result); at 10M it lets t_{surv} halve at matched recall.

7 The Coarse-Cell Corollary

[TODO: With the graph carrying coverage, sweep cell granularity: fine (Kf=262144), medium (65536), coarse (32768). Show medium/coarse WIN because routing evals/query (not coverage) now bind – reversing the “finer is better” intuition. Bracket the knee with four index builds (both finer and coarser lose to the 152-pt-cell knee).]

Effect. At 10M, moving from the fine-cell to the 152-point-cell geometry takes the ScaNN ratio 1.01 $\rightarrow$ 0.85; routing evals/query drop $\sim 12000 \rightarrow 6144$.

8 Joint Cross-Level Partition Optimization (Tree-EM)

[TODO: EM over the hierarchical partition: E-step reassigns to nearest leaf via the beam descent (objective = query-time search), M-step recomputes every level jointly. Cheap-beam E-step. Additive with γ/graph .]

Effect. +0.6–1.0 recall@10 points at 10M at matched probe, QPS-neutral (build-time only).

9 Multi-Assignment (SOAR)

[TODO: Replicate each point into a_0 cells; SOAR spill covers the residual direction of the first cell. A dedup fix (rerank by orig id before the survivor cap) was required to realize it. Raises the routing coverage ceiling.]

Effect. $a_0=2$ reaches OOD coverage parity with the baseline partition; higher a_0 extends the high-recall frontier at a storage/scan cost (quantified in table).

10 Systems Techniques (Streaming)

[TODO: msturing-30M-clustered final_runbook; recall@10 under 1h + 8GB. Three levers: (i) batch-parallel insert (assignment parallelized over the batch, order-preserving, bit-identical); (ii) low-live adaptive p (grow probes so expected candidates $\geq$ threshold when the live set is small); (iii) a parallelization fix (the batched re-rank driver ran chunks serially -> RAYON never engaged). Plus a compliance re-architecture: first-insert-batch cold start + periodic live-set retraining.]

Effect. Batch insert: 30M inserts 1818s $\rightarrow$ 406s ($4.5\times$), recall-identical. Adaptive p : worst step 0.737 $\rightarrow$ 0.944. Parallelization fix: $7.6\times$ at 8 threads. Compliant recall@10 = [pending: final NQ=10000], peak <8GB.

11 Negative Results (Evidence for the Localization)

[TODO: Report each with its measured null/loss:]

- **Query-aware routing** (train cells on the query distribution): *worse* than base-trained (0.76 vs 0.89 @ matched p) — undertrained + the cells where queries are dense don’t distribute base points well.
- **ADC routing** (4-bit-scan the centroids): recall-neutral but *no speed win* — the exact VNNI router is already at the compute floor.
- **Anisotropic/OPQ-rotated coarse codes for rank preservation**: do *not* make coarse codes rank-preserving; reorder depth is set by code bit-rate, and rotation/ η move it < 0.01 . Coverage-via-coarsening collapses (a scann-coverage-matched dpb=50 config yields recall 0.09).
- **RaBitQ**: no ranking edge over PQ at matched bytes.
- **Wider SIMD** (AVX-512 vpermw / 64-wide): scan is memory-bound, not shuffle-bound; $\leq +7\%$, sometimes negative (downclock/cross-lane).
- **Finer cells at matched probe**: lose recall (half the probe fraction).
- **PCA/LeanVec dim reduction on OOD**: near-isotropic embeddings; dead.

These cluster on the scan/code side and all fail; the §5–9 levers cluster on routing/coverage and all succeed — the localization made concrete.

12 Evaluation Methodology

We report *same-hardware ratios*, never cross-machine QPS. All comparisons run on one physical machine; each competitor is measured in the same session, seconds apart, at its own best operating point.

Table 1: OOD text2image, recall@10-matched QPS ratio (ScaNN/ours; < 1 = our engine faster). Same-hardware, tight-pairwise, opponent at official config. [\[pending: 100M row\]](#).

Scale	1-thread	8-thread	recall@10 gate
1M	0.755	0.756	0.905
10M	0.827	0.833	0.901
100M	[pending:]	[pending:]	[pending:]

Table 2: Ablation ladder at 10M (ScaNN/ours ratio; each row adds one lever).

Configuration	ratio
γ only	1.24
+ graph coverage	1.01
+ tree-EM	0.92
+ coarse-cell geometry (knee)	0.85
+ pristine confirmation (20 rounds)	0.83

Protocol. For a target recall ($\text{recall@10} \geq 0.90$), we (i) fix each system’s operating point by an independent sweep, requiring both to clear the recall gate against the *same* exact float ground truth; (ii) alternate the two systems round to round (order flips) so neither owns a load window; (iii) take best-of- k within a round and report the *median of per-round ratios*; (iv) recompute recall from raw ids, never trusting a cached number. On a shared machine, load drift between two builds minutes apart exceeds the effects being measured, so only interleaved same-window ratios and load-independent recall are trusted.

Warm-resident semantics; pristine confirmation. Both systems measured warm and resident (leader-board serving); an early cold-spawn asymmetry is corrected by scoring each engine’s warm block. Headline numbers pause all other first-party jobs (SIGSTOP) and confirm across ≥ 20 rounds.

The baseline at its best. ScaNN uses its *official* big-ANN recipe (e.g. 40k leaves at 10M, anisotropic AH/LUT16, residual quant, bfloat16 re-order) with its own leaves/re-order sweep including corner configs. A win depending on a handicapped baseline is quarantined; we describe one (an under-leaved cached index) and its correction.

What we do not claim. Same-hardware ratios on one (shared) machine, not a standardized-VM submission; ScaNN is the *measured* opponent — gaps to binary-only leaderboard systems are *inference* from the public leaderboard.

13 Results

[TODO: Add per-lever isolation deltas; 8-thread scaling; streaming table (recall, peak mem, wall vs 1h budget). Note the ablation ordering is the actual development arc.]

14 Related Work

[TODO: ScaNN (anisotropic VQ), DiskANN/HNSW (graphs), FAISS IVF-PQ, big-ANN challenge, RaBitQ. Position OOD-specific routing as the gap; distinguish γ from residual quantization (orthogonal) and from query-aware training (shown to fail).]

15 Limitations

[TODO: Single (shared) machine; γ fit per dataset; the coarse-cell corollary is demonstrated on IP/OOD data and may not transfer to in-distribution L_2 ; the gap to binary-only leaderboard systems is inference; filter/sparse tracks untouched; standardized-VM submission is future work and the gating experiment for a leaderboard claim.]